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A comparison of forefoot stiffness in running and running shoe bending stiffness

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Abstract

This study characterizes the stiffness of the human forefoot during running. The forefoot stiffness, defined as the ratio of ground reaction moment to angular deflection of the metatarsophalangeal joint, is measured for subjects running barefoot. The joint deflection is obtained from video data, while the ground reaction moment is obtained from force plate and video data. The experiments show that during push-off, the forefoot stiffness rises sharply and then decreases steadily, showing that the forefoot behaves not as a simple spring, but rather as an active mechanism that exhibits a highly time-dependent stiffness. The forefoot stiffness is compared with the bending stiffness of running shoes. For each of four shoes tested, the shoe stiffness is relatively constant and generally much lower than the mean human forefoot stiffness. Since forefoot stiffness and shoe bending stiffness act in parallel (i.e., are additive), the total forefoot stiffness of the shod foot is dominated by that of the human foot.
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1. Introduction

This study quantifies the stiffness of the human forefoot during running and compares it with the bending stiffness of running shoes. The human stiffness results suggest a need for more sophisticated foot models. The shoe stiffness results pertain to shoe design, since a shoe's relative contribution to forefoot stiffness may affect running performance.

Human joints are often modeled as stiffness elements. Stefanyshyn and Nigg (1998) modeled the ankle joint as an activity-dependent stiffness element, and compared their experimental results with those reported by Davis and Deluca (1995), who measured ankle stiffness during the stance phase of walking. In their study, Stefanyshyn and Nigg defined stiffness as the joint moment divided by the angular displacement.

Isokinetic devices have been used to measure the stiffness of certain joints as a function of time (Hunter and Kearney, 1982). However, dynamic stiffness measurements have not been applied to the forefoot's metatarsophalangeal (MP) joints, which permit the bending of the toes relative to the metatarsal bones during push-off. Dozzi et al. (1989) investigated the MP joint power required during jumping in ballet and found that the muscles near the MP joint provide a great deal of power relative to the joint cross-sectional area.

Early research by Joseph (1954) focused on the first MP joint due to its superior size and strength. This study found that the range of motion of this joint was highly subject-dependent, but in all cases was much larger in dorsiflexion than in plantar flexion. This research further showed that the range of the MP joint in dorsiflexion is significantly greater than that of the interphalangeal joints, suggesting that the portion of the foot distal to the MP joints may be modeled as a single body.

Mann and Hagy (1979) found that the toe muscles are much more active in running than in walking, assisting

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in forward propulsion of the body. A similar study by Bojsen-Moller and Lamoreux (1979) correlated forefoot motion with gait. Hetherington et al. (1990) determined the amount of dorsiflexion required for forward propulsion to take place. Fulford et al. (1989) included inversion and eversion within their study of MP joint dorsiflexion. Lee (1997) and Davies (1999) investigated additional motions of the forefoot.

Foot models have been used for simulation and indirect force determination (Morlock, 1989). Stokes et al. (1979) proposed a two-dimensional MP-inclusive foot model to estimate the forces acting on the MP joint during normal walking. This model was developed further by Salathe and Arangio (1986) to include four rigid bodies. They had concluded that pronation and supination greatly affect the force distribution on the metatarsal heads. Morlock (1989) increased the number of rigid bodies in his MP-inclusive model to six and found that there was no benefit in increasing the degrees of freedom past six. Scott and Winter (1993) created a foot model involving 8 rigid bodies, but limited the degrees of freedom between two connecting bodies to one. Chan and Rudins (1994) used an MP-inclusive model to describe the biomechanics of the foot during running and walking.

The model used in this study is similar to those suggested by Stefanyshyn and Nigg (1997). They quantified the mechanical energy used by the MP joint during running and sprinting and determined that the MP joint was a large energy absorber. Their model considered the distal side of the MP as a single rigid body.

This paper is organized as follows. Section 2 describes the human MP joint experiment and presents the key results. Section 3 presents an experiment evaluating shoe bending stiffness about the MP axis as well as the results of these experiments and their implications. Conclusions are summarized in Section 4.

2. Human forefoot stiffness

In the analysis of the gait cycle, the foot is often modeled as a rigid body. However, this model is not suitable for analyzing the 'push-off' phase of running, in which the MP joint plays a major role (Mann and Hagy, 1979; Morlock, 1989). This section describes an experiment to measure the dynamic stiffness of the MP joint during running.

2.1. Methods

Four male and four female subjects were studied, all between the ages of 20 and 35, with no medical afflictions. All were "moderate" to "active" in their sports activities. Table 1 lists their height and weight. Informed consent was obtained from all subjects.

Reflective ball markers, 1.2 cm in diameter, were placed on the right foot so that the motion of foot segments can be tracked. Fig. 1 shows the placements of these markers. Markers 6, 8, and 9 define the forefoot, as in Lee (1997). Markers 3, 4, and 5 define the rearfoot, and were used to determine the relative rotation of the forefoot.

Six high speed Falcon 240 Video Cameras (Motion Analysis, Santa Rosa, California) with 6.0 mm Computar lenses were placed at heights and angles such that a minimum of four cameras see any marker during the stance phase of a recorded running stride. The cameras are set to a frame rate of 240 Hz and are calibrated using a cube and a moving wand.

A Kistler force plate measured the ground-reaction force. Since the force plate has a load cell on each of its four corners, it yields the magnitude, direction, and point of application of the resultant force (Nigg and Herzog, 1994). Before testing each subject, the positions of the four load cells were calibrated by placing a video marker at each. For accuracy, ground reaction force (GRF) positions were computed from individual load cell data instead of using the Kistler hardware that performs this function. To match the video frame rate,

Table 1
Subjects

Name	Sex	Height (cm)	Weight (kg)
Subject 1M	Male	183	84
Subject 2M	Male	170	70
Subject 3M	Male	168	74
Subject 4M	Male	179	82
Subject 1F	Female	165	61
Subject 2F	Female	168	64
Subject 3F	Female	172	61
Subject 4F	Female	162	58

Markers	Location Name
1	Medial Ankle
2	Lateral Ankle
	(not used in study)
3	Proximal Calcaneus
4	Distal Calcaneus
5	Posterior Calcaneus
6	5th Metatarsal Head
7	2nd Metatarsal Head
	(not used in study)
8	2nd Phalange
9	1st Metatarsal Head

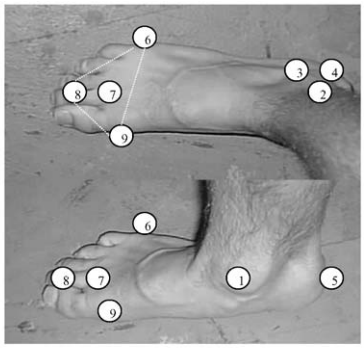


Fig. 1. Marker locations.

force data was down-sampled to 240 Hz from 2400 Hz by averaging each consecutive set of ten points.

Each subject ran barefoot at 4 (±.2) meters per second and landed with his right foot near the center of the force plate. Data collection was triggered by a motion sensor. Ten trials were performed for each of the eight subjects.

2.2. Results and discussion

In this section, the measured flexion of the forefoot and the GRF are described with respect to a foot coordinate system. The moment about the MP joint and its effective stiffness are then computed. The results presented for each subject are the mean results from ten trials. One male and one female subject were removed from the results, as they were not heel-toe runners and hence could not be compared easily with the other subjects.

Fig. 2 shows the foot coordinate system, which is considered fixed to the rearfoot. The Z-axis coincides with the MP joint axis, which is defined as the line segment joining the first and fifth MP joint heads, represented by markers 9 and 6 (respectively) in Fig. 1. The X-axis is perpendicular to the Z-axis and is horizontal when the foot lies flat on the ground. This foot position will be referred to as the neutral stance. The Y-axis completes a left-handed coordinate system and is nearly vertical in the neutral stance.

Fig. 3 shows, for the six heel-toe subjects, the GRF component along the foot Y-axis versus percentage push-off. The forces are negative along the Y-axis, as they are considered to be applied *to* the ground *by* the foot. Percentage push-off is defined as the normalized time during push-off, which commences at the moment of heel lift and ends at toe-off. The GRF is not plotted for percentage push-off near zero because the GRF acts primarily behind the MP joint, giving a negative measured moment. This is not the actual *bending* moment since it does not cause bending of the joint.

Fig. 4 shows the corresponding X-axis GRF. Since only the push-off phase is shown, this force is primarily

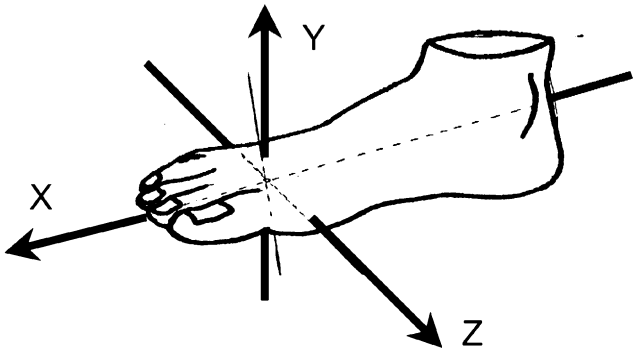


Fig. 2. Foot coordinate system.

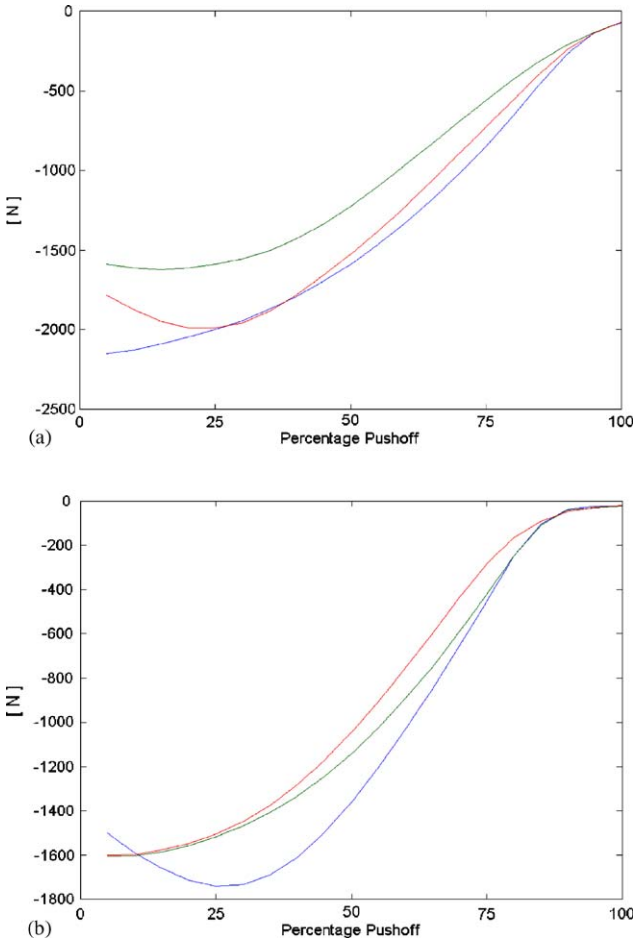


Fig. 3. (a) Y-axis force (men) and (b) Y-axis force (women).

negative (producing acceleration) and does not integrate to zero. This force contributes to the bending moment about the MP axis (Z-axis), but contributes much less than the Y-axis force component.

Fig. 5 shows the bending moment about the MP axis (Z-axis) produced by these forces. This moment is computed as the product of the resultant GRF and its moment arm, which is found by comparing the line of action of the force (from force-plate data) with the MP joint position (from video data). A negative value of bending moment indicates dorsiflexion of the joint. Prior to push-off, the moment is positive due to forces acting behind the MP joint. However, such positive moments are not *bending* moments—only forces forward of the joint can cause bending (joint flexion). Such forces may be a source of error during push-off: a vertical force acting on the sesamoid behind the MP joint adds to the *computed* bending moment, but not to the *actual* bending moment. This is shown in Fig. 6. Because the actual bending moment is negative in sign, this error causes a conservative measurement of the bending moment magnitude and hence a conservative forefoot stiffness. The curves in Figs. 5 and 7 do not

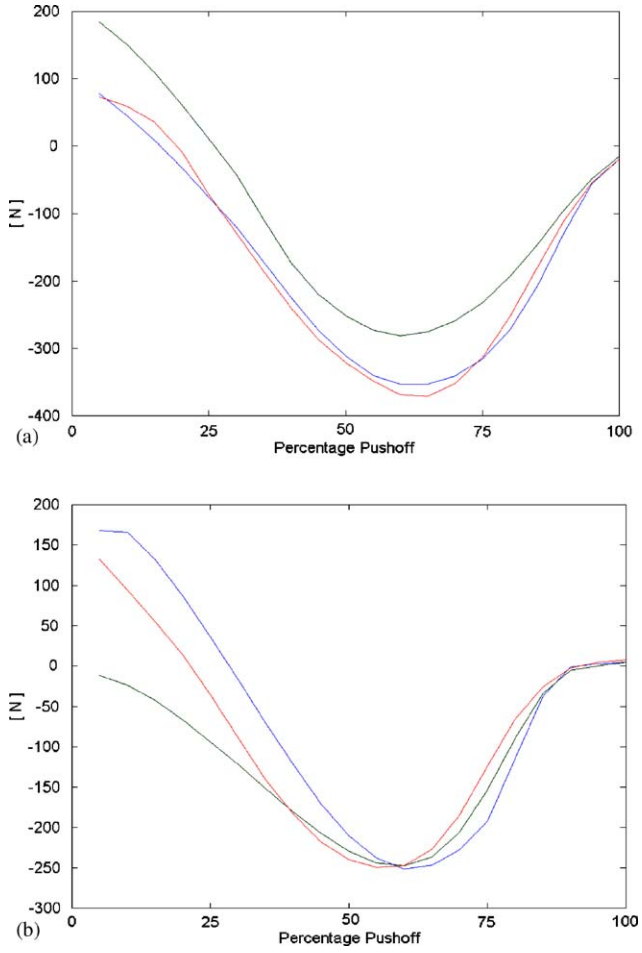


Fig. 4. (a) X-axis force (men) and (b) X-axis force (women).

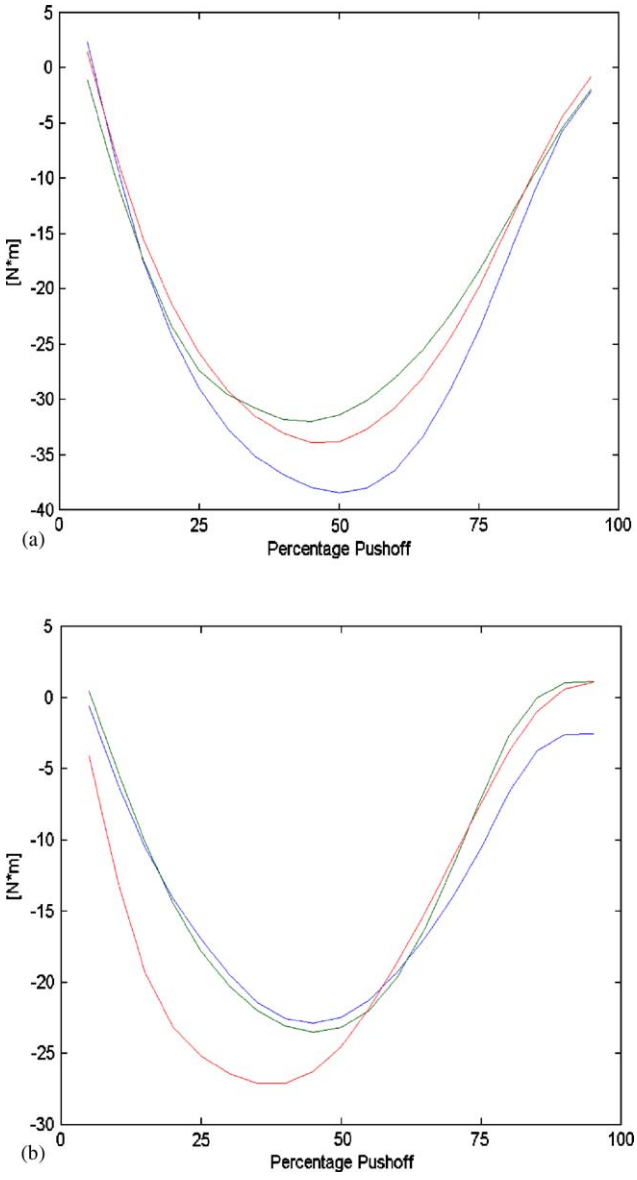


Fig. 5. (a) Moment about Z-axis (men) and (b) moment about Z-axis (women).

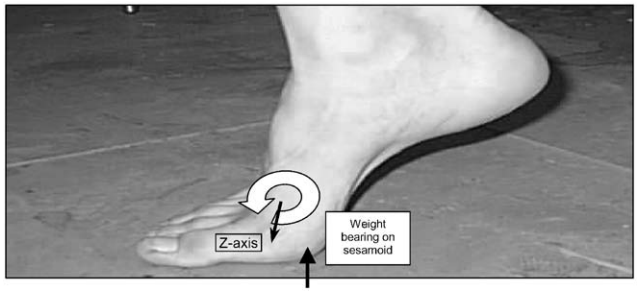


Fig. 6. Bending moment error produced by sesamoid force behind MP joint.

continue to 100% push-off, as the data becomes unreliable near toe-off.

Fig. 7 shows the flexion of the MP joint, with negative values representing dorsiflexion (relative to the neutral stance). Dorsiflexion first increases as the rearfoot rotates forward onto the forefoot and then decreases just before toe-off.

Fig. 8 shows mean bending moment versus mean dorsiflexion angle, with the mean taken over the three heel-toe subjects in each gender group. The neutral stance corresponds to zero flexion. Time is not shown, but progresses in a clockwise direction from the bottom left of the plot. It can be seen that the dorsiflexion angle is not a function of moment alone, as there are two values of angle for each value of moment, with the higher value occurring later in time in Figs. 13 and 14. Nor is moment a function of angle alone, as angles above 35 degrees (for men) do not correspond to unique moments. Hence, both of these variables depend explicitly on time.

Fig. 9 shows forefoot stiffness versus dorsiflexion angle. Time progresses in the clockwise direction starting from the lower left of the plot. As in Stefanyshyn and Nigg (1998), stiffness is defined as the bending moment divided by the angle of flexion, where these are calculated as described above. The stiffness varies

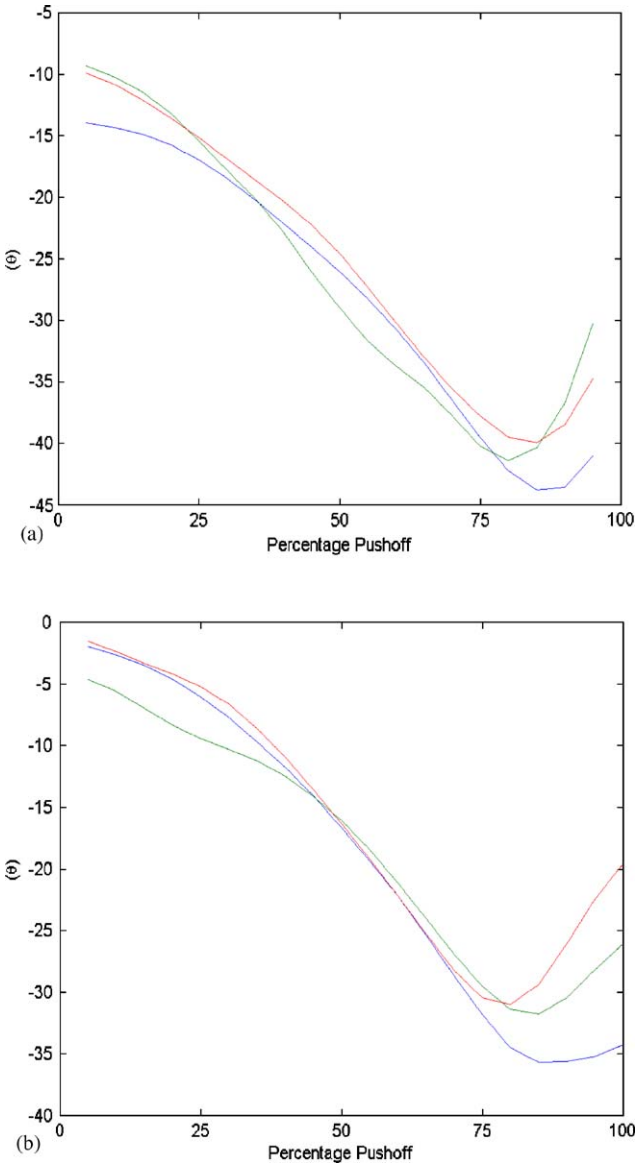


Fig. 7. (a) Angle about Z-axis (men) and (b) angle about Z-axis (women).

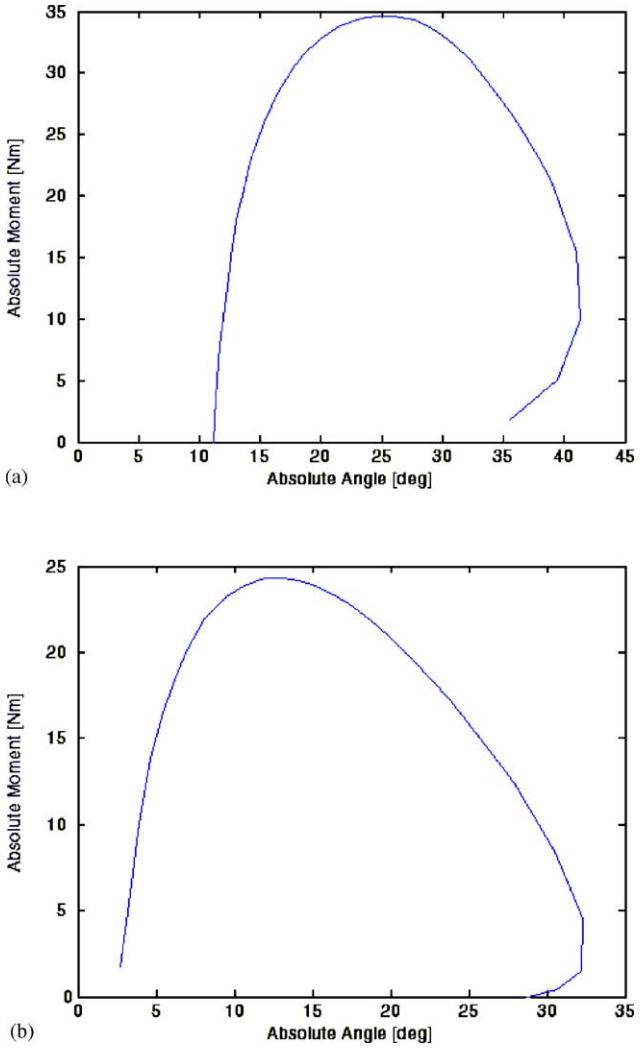


Fig. 8. (a) Moment vs. Flexion Angle (men) and (b) moment vs. Flexion Angle (women).

greatly with the angle of flexion and is not unique at angles above 35 degrees (men). Thus, the stiffness is also explicitly dependent on time.

The intra-subject repeatability of the experiment is indicated in Fig. 10, which shows the moment data for Subject #1 over ten trials. For each subject, the standard deviation of the moment (over the ten trials) was found as a function of time. The mean value of this deviation (over time) was then computed and normalized by the range of the moment values. The resulting 'intra-subject deviation' ranged from 5.5% for Subject #5 (female) to 10.5% for Subject #4 (male). The standard deviation of the moment *between* subjects was found in the same way by comparing the mean moment curves of subjects. The resulting 'inter-subject deviation' in moment was 5.9% in the male group and 7.1% in the female group.

As mentioned previously, a possible source of error is the false bending moment contributed by weight-bearing forces on the sesamoid behind the MP joint, as shown in Fig. 6. However, Stefanyshyn and Nigg (1997) showed that this effect results in only a small underestimation of the bending moment, as the sesamoid is only partially in the rearfoot.

3. Shoe bending stiffness

In this section, the effective shoe bending stiffness about the MP joint is found for four running shoes and compared with the forefoot stiffness determined in Section 2. The shoe bending stiffness acts in parallel to the forefoot stiffness, thus adding to it. Although runners seem to prefer flexible footwear, the need for flexibility within the sole of the shoe has yet to be proven

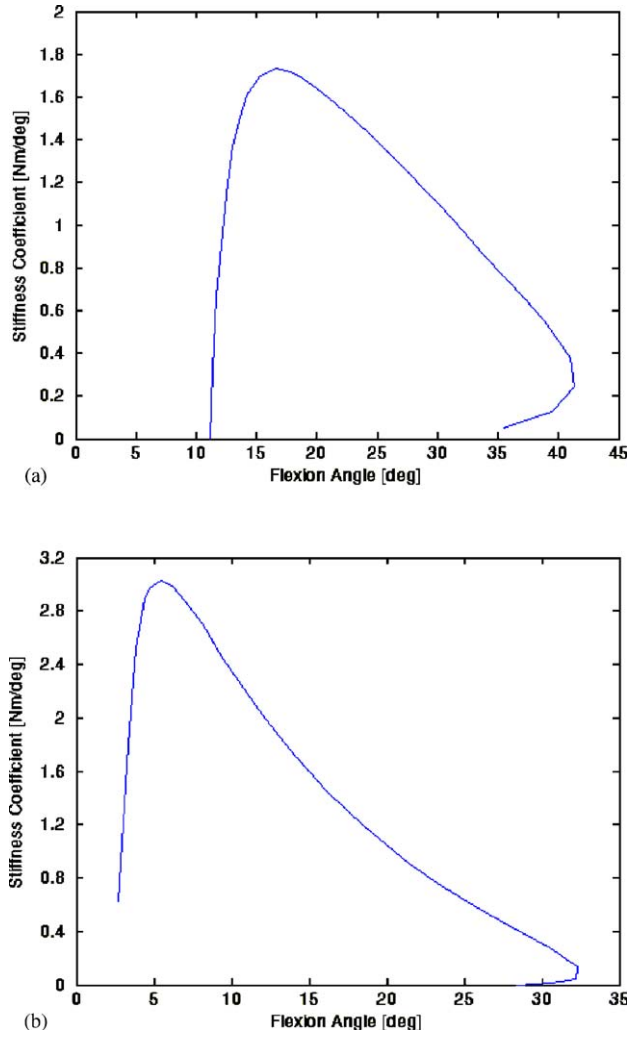


Fig. 9. (a) Stiffness vs. Flexion Angle (men) and (b) stiffness vs. Flexion Angle (women).

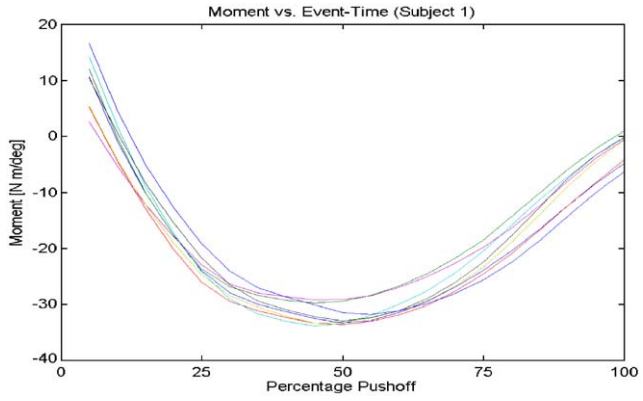


Fig. 10. Intra-subject repeatability in MP bending moment (subject #1).

scientifically. A sport shoe is designed such that the shoe can bend at least 30 degrees at a point just behind the metatarsal heads with relative ease. Research by Stefanyshyn and Nigg (1997) indicates that shoe bending stiffness at the MP joint may be too low.

3.1. Methods

Like the foot stiffness experiment of Section 2, the shoe stiffness was found from dynamic measurements. Quasi-static measurements, included for comparison, gave lower stiffness values, possibly because air trapped in the porous material of the sole has time to escape, thus reducing its contribution to stiffness. A key difference between the foot and shoe stiffness tests is that the forefoot bends about a fairly well-defined axis (the MP joint), whereas shoe bending is distributed over some area on the sole. However, this area is effectively restricted by the foot, to which the shoe must conform to some extent. This restriction is effected by anchoring a rigid last to the inner sole of the shoe.

The experimental setup is shown in Fig. 11. The tested running shoes, described in Table 2, cover a range of sole designs and wear conditions. The shoe is held in place by a rigid, steel rearfoot last bolted to a table. Vertical forces are applied to the forefoot by an MTS machine (model 510.10C) and measured by an MTS axial force transducer (model 661.19E-01, 5000.00 N). A flat, rigid steel plate is anchored to the sole of each shoe using a C-clamp at point A, as shown in Figs. 11 and 12.

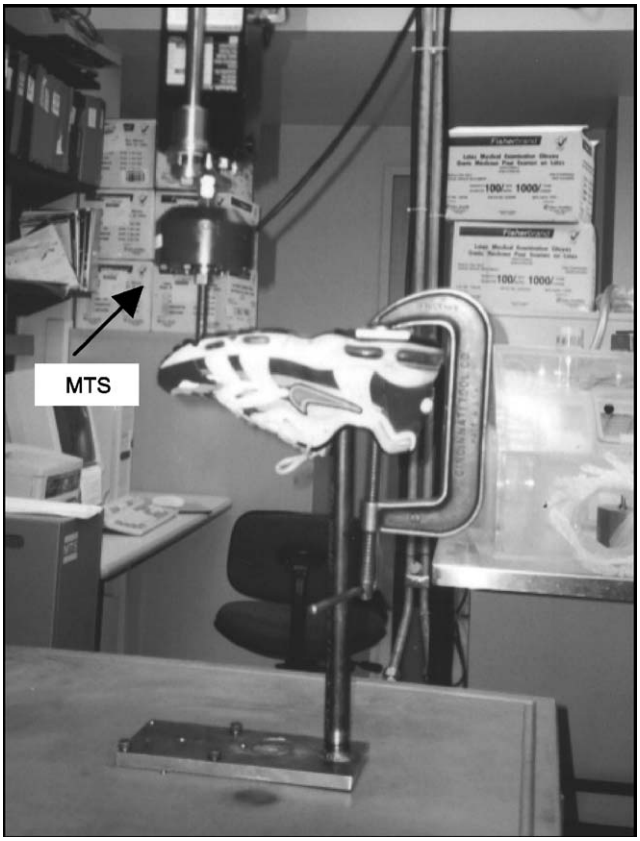


Fig. 11. Experimental setup for shoe stiffness experiments.

Table 2
Running shoes tested

Shoe	Make	Model	Model #	Sole construction	Size	Condition
A	Nike	Air Max	104076 131	Air-filled (toe & heel)	M10	Medium Wear
B	New Balance	N1022	M1022NC	Foam	M10.5	New
C	Nike	Air Max Octane	302415 001	Foam	M10.5	Medium Wear
D	Nike	Air Max	104111 071	Air-filled (full sole)	M10	Heavy Wear

Padding, placed between the plate and the shoe tongue at point B, prevents movement of the plate. Shoe lengths ranged from 30 cm to 32.5 cm, while distance d in Fig. 12 varied from 5.5 cm to 7.5 cm among trials.

For each trial the MTS probe is lowered vertically at a constant speed through a height difference of 20 mm. Each shoe was tested at speeds of 2.5 mm/s (quasi-static), 100 and 300 mm/s (top speed of MTS). The vertical force measured by the load cell is plotted versus time in Figs. 13 and 14. The shoe deflection is also plotted versus time, with the vertical axis representing millimeters instead of newtons. The MTS logged data every 0.004 s. Fig. 13 shows force and deflection versus time for both the down and up phase of the MTS at a speed of 2.5 mm/s. After descending at 2.5 mm/s, the MTS probe remains at constant height for 5 s and then proceeds upwards. Fig. 14 shows a similar plot for only the loading phase at a speed of 300 mm/s.

Visible bending of the sole occurred between the two parallel lines marked in Fig. 15. Distance p in the figure remained approximately constant at 2 cm for all trials. The bending moment is obtained as the product of the true force F_{True} and the true moment arm d_2. These are found from the measured vertical force F_{MTS} and the horizontal moment arm d_1, as shown in Fig. 16. Distance d_1 is measured from the point of application of the actuator to the point midway between the parallel lines in Fig. 15.

3.2. Results and discussion

Similar to the forefoot stiffness of Section 2.2, the shoe bending stiffness is computed as the ratio of bending moment to bending angle. In Figs. 17, 18 and 19, the stiffness of the four shoes is plotted against the change in bending angle and is compared with the forefoot stiffness curves of Fig. 8. The shoe stiffness curves are given for both the loading and unloading phases of the trials. In each case, the stiffness associated with the loading phase is greater than that of the unloading phase.

As expected, the shoe stiffness obtained from quasi-static measurements (Fig. 17) was lower than that obtained dynamically (Figs. 18 and 19), and a smaller difference between loading and unloading stiffness (i.e., hysteresis) was observed. The stiffness of each shoe is relatively constant compared to human forefoot stiffness, but varies considerably among the four shoes. For most of the bending range, the forefoot stiffness is much larger than the bending stiffness of the shoes. Hence, the combined stiffness of the foot and shoe is dominated by that of the foot, which suggests that typical variations in shoe bending stiffness may not be critical to running

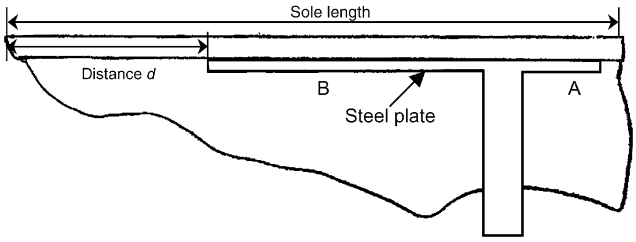


Fig. 12. Shoe and last.

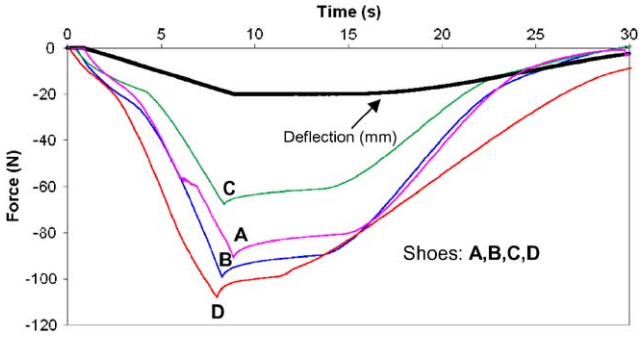


Fig. 13. Force and deflection vs. time at 2.5 mm/s (loading and unloading).

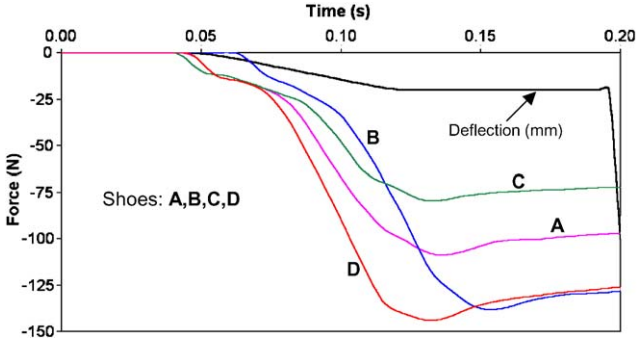


Fig. 14. Force and deflection vs. time at 300 mm/s (loading only).

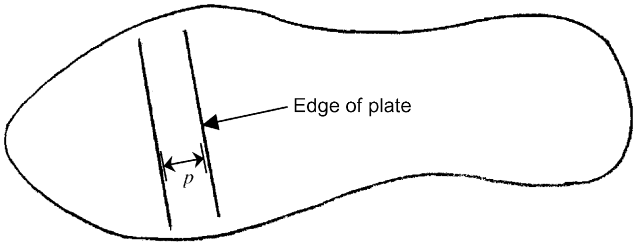


Fig. 15. Shoe sole.

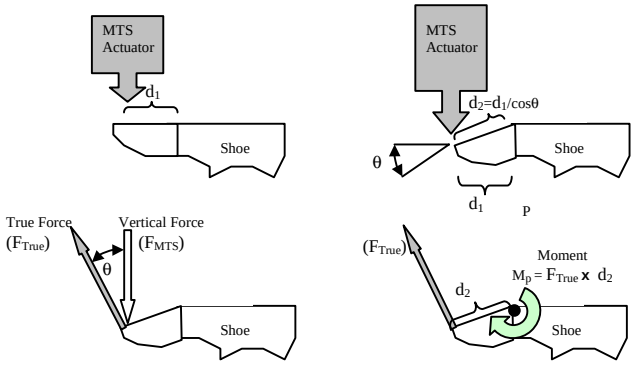


Fig. 16. Computation of bending moment.

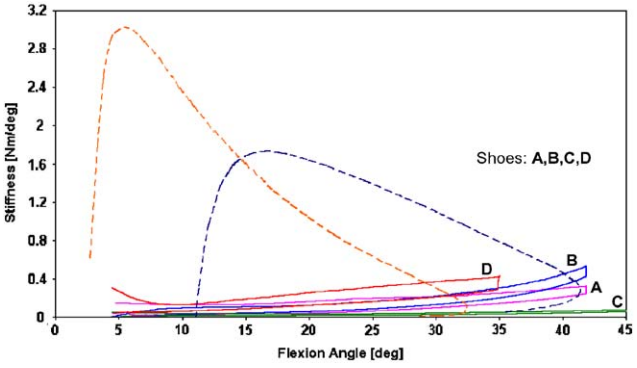


Fig. 17. Comparison of shoe stiffness and foot stiffness at 100 mm/s.

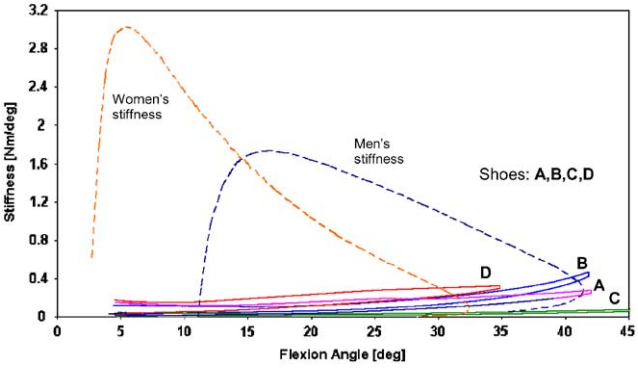


Fig. 18. Comparison of shoe stiffness and foot stiffness at 2.5 mm/s.

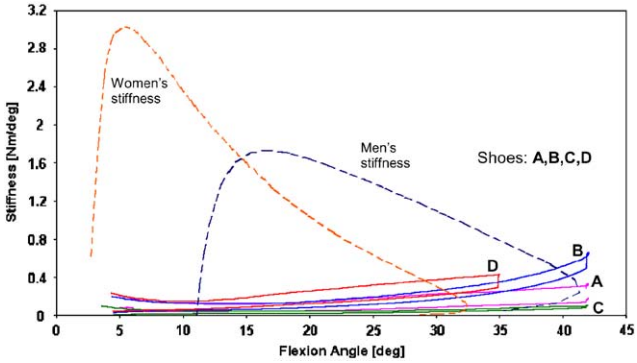


Fig. 19. Comparison of shoe stiffness and foot stiffness at 300 mm/s.

performance. On the other hand, the smaller forefoot stiffness near toe-off suggests that shoe bending stiffness may be of some importance in toe-off.

4. Summary and conclusion

The stiffness of the human forefoot was measured for six subjects running barefoot. It was shown that as the joint deflects, the effective forefoot stiffness rises sharply and then decreases steadily. This indicates that the forefoot does not behave as a simple spring, but rather as an active mechanism with a highly time-dependent stiffness. The forefoot stiffness was compared with running shoe bending stiffness about the same axis. For each of the four shoes tested, the shoe stiffness was much smaller than the human forefoot stiffness.

This study reveals the dynamic nature of the forefoot stiffness, indicating that a passive stiffness model is inadequate. It does not provide a new forefoot model, but suggests that for a given activity, the forefoot moment is explicitly dependent on time. For example, the moment may depend jointly on time and the angle of flexion (which itself depends on time for a given activity, subject, surface, etc.).

Since forefoot stiffness and shoe bending stiffness act in parallel (i.e., are additive), the total forefoot stiffness of the shod foot is dominated by that of the human foot. This suggests that typical variations in the bending stiffness of running shoes are unlikely to have a significant effect on running performance.

An important limitation of this study is that the forefoot stiffness measurements were based on only six subjects, who may not be representative of the general population. The small dorsiflexion angles measured in this study suggest that these subjects may have a significant functional hallux limitus condition. It is recommended that future studies consider a larger population of subjects and evaluate their MP joint mobility prior to testing.

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